



Urban morphology and daytime surface heat production in a rapidly transforming tropical megacity: Evidence from Ho Chi Minh City, Viet Nam

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ABSTRACT

Rapid urbanisation in tropical delta cities is intensifying urban heat island effects. Yet analyses often treat warming as a generic land-cover effect rather than an outcome of distinct urban development fabrics and trajectories. This paper examines how urban morphological structures shaped dry-season land surface temperature (LST) in Ho Chi Minh City (HCMC) between 2010 and 2020 using a block-based Urban Structure Type (UST) framework that captures built form, function, and green–blue spatial structure. Dry-season Landsat LST data were analysed to assess surface thermal change, typology-level contrasts, landscape–temperature relationships, and typology-specific variable importance. Results show a citywide upward shift in dry-season LST and an outward spread of higher-temperature areas from the historical core into industrial corridors and peri-urban growth zones. A persistent thermal hierarchy is evident across both years, with the highest LST observed in industrial/logistics and dense shophouse fabrics, and lower LST in less compact built-up patches and vegetation- or water-dominated surfaces. Grid-based change analysis links the expansion of industrial, commercial/special-use, and under-construction fabrics to increased LST, whereas losses of forest and cropland are associated with warming. Landscape metrics indicate that composition and dominance, especially the areal share of a typology (PLAND), relate to LST more consistently than fragmentation metrics, while Random Forest results show that variable importance is typology-specific. The UST approach makes these findings planning-relevant by identifying hotspot-producing urban fabrics and potential surface-heat lock-in, creating a practical bridge from Earth observation to typology-specific surface-heat mitigation in rapidly transforming tropical megacities.

1. Introduction

Urban heat is not merely a biophysical by-product of urbanisation, but a spatial imprint of how cities expand, densify, and reorganise land, infrastructure, and ecological systems. By converting vegetated and permeable surfaces into impervious built-up areas, urban growth alters surface energy balances through reduced evapotranspiration, changed radiative properties, increased heat storage

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in materials, and modified turbulent exchanges between the land surface and the atmosphere (Arnfield, 2003; Oke, 1982; Weng, 2009). These transformations materialise as measurable changes in land surface temperature (LST), a widely used indicator of surface thermal conditions and landscape restructuring (Diem et al., 2024b; Voogt and Oke, 2003). Decades of research show that cities commonly exhibit higher surface temperatures than their surrounding countryside (Oke et al., 2017), while multi-city and long-term remote-sensing studies indicate that surface urban heat island (SUHI) intensity has generally strengthened in recent decades (Liu et al., 2022; Peng et al., 2012; Si et al., 2022). Recent global analyses further suggest that this intensification is occurring particularly rapidly in low- and lower-middle-income settings, where rapid urbanisation often converges with constrained planning capacity, infrastructure limitations, and insufficient adaptive resources (Yuan et al., 2025).

At the same time, LST must be interpreted carefully. It captures spatial differences in surface thermal conditions, but does not directly measure human heat stress, thermal comfort, or health risk. Human heat impacts also depend on near-surface air temperature, humidity, radiation, wind, exposure duration, nocturnal cooling, indoor conditions, socio-economic vulnerability, and adaptive capacity. Previous studies show that lower-income and marginalised populations are often disproportionately exposed to elevated urban temperatures and associated heat burdens, and that such disparities may intensify under climate change scenarios (Chakraborty et al., 2019; Dodman et al., 2022). In Ho Chi Minh City, extreme temperatures have also been associated with elevated mortality and hospital admissions (Dang et al., 2019, 2018; Phung et al., 2017). Accordingly, this study uses LST to examine surface thermal patterns and heat-exposure potential associated with urban morphology, rather than to provide a comprehensive heat-risk assessment.

Urban heat is therefore both physically structured and socially mediated. Critical heat studies emphasise that vulnerability is shaped by intersecting inequalities in housing quality, access to infrastructure, energy affordability, planning systems, shade and vegetation, and adaptive capacity, rather than by exposure alone (Iqbal et al., 2026; Nguyen et al., 2022; Peker and Street, 2025; Tschakert et al., 2025; Tschakert and Karthikeyan, 2025). These wider vulnerability dimensions are not directly assessed here, but they underline the planning relevance of examining how urban transformation reshapes surface thermal conditions. The narrower gap addressed in this study concerns typology-based urban heat research, which often remains static and therefore provides limited insight into how morphological change reconfigures surface thermal conditions in rapidly transforming cities.

Despite growing scientific consensus on urban heat as a major climate concern, it remains unevenly institutionalised in planning practice, particularly in rapidly urbanising regions. Reviews of urban climate action planning show that heat, flooding, and compound risks are widely acknowledged in policy discourse, yet cities often lack the institutional capacity, coherent regulatory instruments, and evidence-based frameworks required to translate climate knowledge into spatially grounded action (Aboagye and Sharifi, 2026; Berrang-Ford et al., 2021; Nguyen et al., 2026; Olazabal and Ruiz De Gopegui, 2021). Even where detailed urban climate analyses exist, thermal evidence is often weakly embedded in statutory planning instruments and development control (Eliasson, 2000). As a result, urban heat is frequently treated as an external climate stressor rather than as an outcome shaped by planning decisions, infrastructure sequencing, land conversion, and everyday development trajectories.

Analytically, the Local Climate Zone (LCZ) framework provides an important harmonised basis for comparing urban thermal landscapes, demonstrating that SUHI intensity is structured by built form, surface cover, materials, and human activity rather than by land cover alone (Stewart and Oke, 2012). Multi-city evidence shows that compact, high-rise, and dense commercial LCZs often exhibit high daytime SUHI intensity, whereas open and vegetated classes tend to be cooler, though these differences vary with background climate and local development context (Bechtel et al., 2019). Recent studies further show that growth morphology and spatial disorder can amplify LST trends in developing coastal megacities (Halder and Garg, 2025), while LCZ-based knowledge still faces challenges of institutional uptake, scale mismatch, and translation into planning practice (Yang et al., 2025). Evidence also cautions that zone-average temperatures may mask critical hotspots if maximum LST and within-zone heterogeneity are not considered (He et al., 2024).

Urban Structure Types (USTs) provide a complementary lens for this purpose. USTs translate spatial configurations into interpretable urban morphologies shaped by distinct growth pathways, capturing variation in building forms, plot coverage, street configurations, vegetation distribution, and functional land-use mixtures more directly than conventional land-use and land-cover categories do. Because planning instruments often regulate typology-like attributes such as building form, plot coverage, street sections, land-use function, and green-blue structure, USTs provide a practical bridge between satellite-derived thermal diagnostics and planning-relevant urban forms (Downes and Storch, 2014). This approach is especially relevant because urban form and spatial configuration, including density, compactness, fragmentation, and aggregation, can modulate surface heat patterns as strongly as surface composition, with relationships that are scale- and context-dependent (Dugord et al., 2014; Forman, 1995; Liu and Weng, 2009; McGarigal et al., 2002; Rahimi et al., 2021; Yue et al., 2019; Zhang et al., 2015). Recent work further suggests that horizontal densification can intensify LST, while vertical form may reduce surface thermal load through shading and canyon effects, with vegetation and water moderating thermal extremes (Jeon et al., 2023). Yet UST-based thermal analyses remain rare in Southeast Asia, and HCMC has not yet been examined through a typology-change lens that links development trajectories to surface heat production.

HCMC provides a critical setting for this analysis. Viet Nam's national adaptation agenda recognises extreme heat as a growing concern, including more frequent very hot days, often defined as days with maximum temperatures $\geq 35^\circ\text{C}$. Since the 1990s, and accelerating over the last two decades, rapid peri-urbanisation and population growth have driven extensive conversion of cropland and other peri-urban landscapes into residential, industrial, and production spaces, alongside broader economic restructuring (Downes et al., 2024; Kontgis et al., 2014; Schaefer and Thinh, 2019). Prior studies report increased LST in high-density and industrial zones, underscoring the surface thermal consequences of rapid and uneven development in HCMC (Doan et al., 2016; Katzschner et al., 2023; Son et al., 2017; Thi Van and Duong Xuan Bao, 2010). Comparable trajectories across Southeast Asian megacities suggest that surface warming is jointly shaped by rapid urban expansion, vegetation loss, impervious surface growth, and broader climatic change (Estoque et al., 2017; Nguyen et al., 2022).

Against this background, this study examines how urban growth and morphological change shaped dry-season surface thermal

conditions in HCMC between 2010 and 2020 by integrating satellite-derived LST with a UST typology and landscape configuration metrics. The paper makes three contributions. First, it advances a process-based account of surface heat production by linking morphological transitions, rather than static land cover, to changes in LST across 16 UST classes. Second, it provides a morphology-sensitive diagnosis of urban thermal structure by testing how landscape composition and configuration relate to LST within and between typologies. Third, it translates these insights into planning-relevant evidence by specifying typology-targeted leverage points for surface-heat mitigation that are legible to zoning, development control, and urban climate adaptation practice. The remainder of the paper introduces the study area and datasets, outlines the USTs, LST, landscape metrics, and Random Forest methods, presents results on spatiotemporal LST patterns and typology-specific drivers, and concludes with implications for climate-responsive planning in rapidly urbanising tropical megacities.

2. Methodology

2.1. Study area

HCMC is an emerging megacity in southern Viet Nam, covering approximately 2095 km² and located between 10°10′–10°38′ N and 106°22′–106°54′ E (Fig. 1). Administratively, the city comprises 24 districts spanning dense urban cores, industrial zones, peri-urban expansion areas, and remnant agricultural and low-lying wetland landscapes. Since the economic reforms initiated under the "Đổi Mới" policy in the late 1980s, HCMC has experienced sustained population growth, industrial expansion, and rapid peri-urban land conversion, supported by deepening integration into regional and global production networks, including export-processing zones, high-tech parks, and logistics corridors (Leitold et al., 2021; Nguyen et al., 2016; Schmidt et al., 2013; Storch and Downes, 2011). The official population exceeded 9 million in 2020 (General Statistics Office, 2020), while the functional metropolitan population is substantially higher due to unregistered residents and daily commuting flows (Downes et al., 2016; Downes and Storch, 2014).

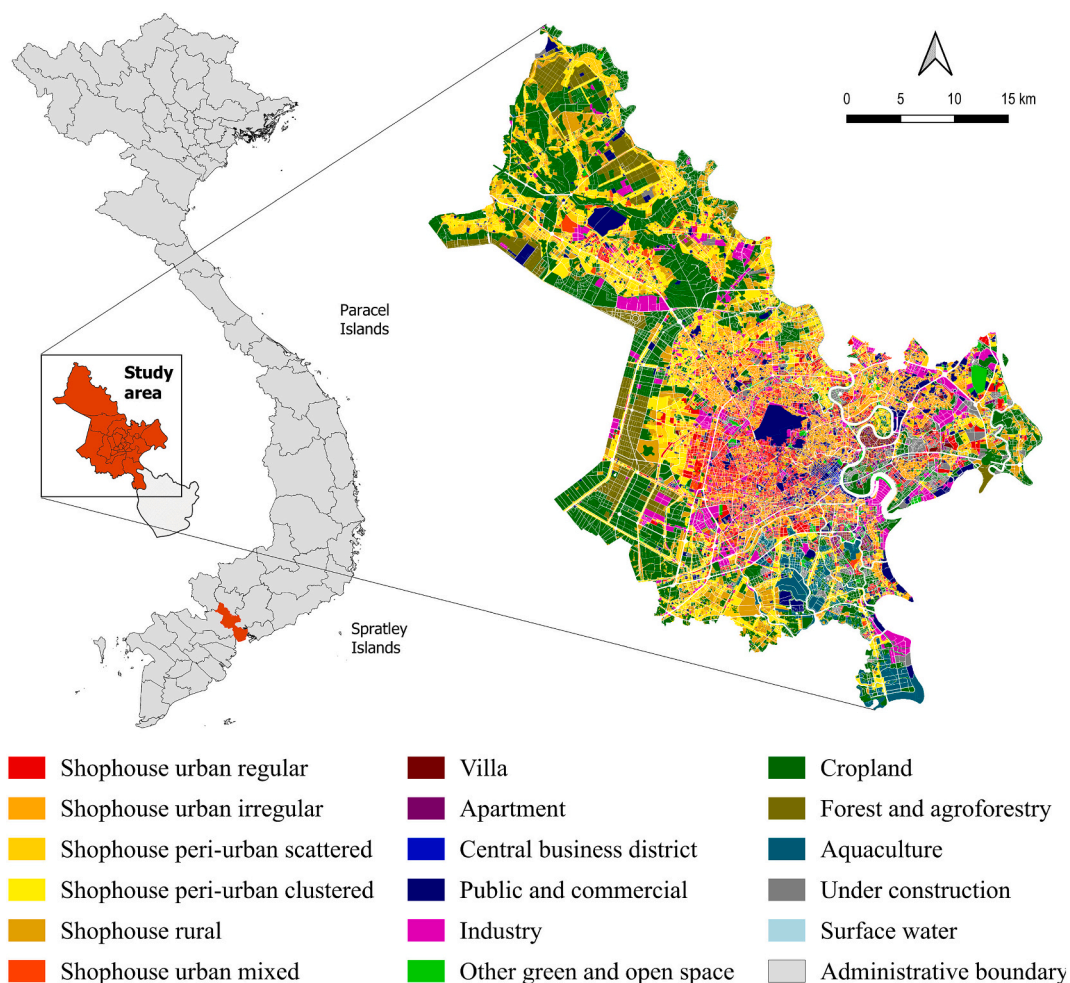


Fig. 1. Location of Ho Chi Minh City within Southern Viet Nam and the spatial distribution of USTs in 2020.

Climatically, HCMC lies within the tropical wet–dry (Aw) climate zone according to the Köppen–Geiger classification. Temperatures remain high year-round, with mean annual temperatures of approximately 27–28 °C and peak monthly averages exceeding 30 °C during the late dry season (March–May). The climate is characterised by a rainy season from May to October and a dry season from November to April. Average annual rainfall is approximately 1800 mm, and high solar radiation contributes to a high baseline thermal load, which can intensify surface heat effects associated with dense urban development.

Located in the lower Dong Nai–Saigon River basin, agricultural, wetland, and semi-rural landscapes have been progressively transformed into dense residential, industrial, and commercial areas, producing new urban morphologies that embed interacting surface-thermal and flood-exposure concerns shaped by land conversion and terrain constraints (Katzschner et al., 2023; Scheiber et al., 2025; Tu et al., 2026, 2025, 2024). These dynamics make HCMC a critical setting for examining how urban growth pathways shape spatially differentiated surface thermal conditions. Can Gio District (approximately 704 km²) was excluded from the analysis due to persistent cloud contamination in available scenes, limited availability of usable thermal observations, and its predominantly mangrove and low-density land cover, which is not representative of the city's dominant urbanisation trajectories.

2.2. Data collection and preprocessing

2.2.1. Satellite-based LST data

Daytime LST was derived from Landsat 5 Thematic Mapper (TM) imagery (2009–2010) and Landsat 8 OLI/TIRS imagery (2020) to assess changes in surface thermal conditions across HCMC.

All Landsat scenes were geometrically co-registered to ensure spatial consistency across years. For the reflective bands used to compute vegetation indices (i.e., red and near-infrared bands), atmospheric effects were reduced using the Dark Object Subtraction (DOS) method (Chander et al., 2009; Masek et al., 2006). LST for circa 2010 and 2020 was derived from Landsat thermal infrared (TIR) bands using a standard single-channel retrieval workflow. Digital numbers were converted to at-sensor radiance using sensor-specific calibration parameters and subsequently transformed to brightness temperature using the inverse Planck function (Chander et al., 2009; Li et al., 2013). Land surface emissivity was estimated using a vegetation index-based (NDVI, normalised difference vegetation index) approach to account for surface heterogeneity and vegetation cover (Sobrino et al., 2004). Final LST was then obtained by applying emissivity correction to brightness temperature, following established procedures widely used in urban thermal studies (Weng et al., 2004).

For each period, all single LST scenes captured during dry-season observations (February–May) were composited using median pixel values. This compositing approach reduces the influence of cloud contamination, residual haze, and short-term atmospheric variability while preserving the seasonal thermal signal most relevant to urban heat analysis. Although compositing choices can influence thermal outputs under different atmospheric regimes (Xu et al., 2025; Weng et al., 2004), the median composite provides a robust, outlier-resistant representation of prevailing dry-season surface temperatures across multiple scenes.

To address potential scale effects, LST-urban form relationships were assessed across multiple spatial extents, consistent with evidence that urban heterogeneity-LST relationships can be scale-dependent (Rahimi et al., 2021).

2.2.2. Urban structure type (UST) data

USTs were derived from an existing validated process-typological classification framework developed by Downes and colleagues, which distinguishes locally specific building archetypes mapped to official land-use geometries (Downes, 2019; Downes et al., 2024, 2016). For this analysis, the original fine-grained typology was aggregated into 16 dominant urban and land-cover classes capturing key differences in built form and land-use function. These include apartment areas, the central business district (CBD), shophouse typologies, industrial and public-commercial zones, villas, peri-urban and rural settlements, and non-built classes such as agricultural land, aquaculture, forested areas, and areas under construction.

This aggregation preserves meaningful variation in building form, surface characteristics, and development intensity, while providing a consistent spatial framework for evaluating associations between urban structure and LST. UST transitions between 2010 and 2020 have been examined in detail elsewhere (Downes et al., 2024). In the present study, USTs are employed as an explanatory spatial framework rather than to re-evaluate urbanisation pathways or quantify land-use change. The mapped UST configuration, therefore, provides the morphological context for analysing spatial variation in LST across urban forms.

2.2.3. Spatial units and analytical framework

Daytime LST was initially derived at the Landsat-pixel level. The UST dataset consists of mapped blocks, each assigned to one of the 16 UST classes. For comparing LST among USTs, pixel-level LST values were spatially averaged within each UST block, and these block-level mean values constituted the observations used in the ANOVA and Tukey HSD tests. For the change and landscape-metric analyses, the UST and LST datasets were summarised within a common 500 × 500 m grid. Each grid cell constituted the statistical observation for the Δ UST– Δ LST regression, Pearson correlation, and Random Forest analyses. Within each grid cell, contiguous areas belonging to the same UST class were treated as patches for calculating class-level landscape metrics; these patches were metric components rather than independent statistical observations. District-level summaries were used only for descriptive spatial interpretation and discussion of planning relevance.

2.3. Landscape indices calculation

Landscape structure was quantified using three groups of class-level landscape indices: (i) area and edge metrics, (ii) shape metrics,

and (iii) aggregation metrics. These indices are widely applied in landscape ecology and urban spatial analysis to characterise urban form and its influence on environmental processes, including surface thermal behaviour (McGarigal et al., 2002; Turner et al., 2001). Definitions and formulas for each metric are provided in Table 1.

Area and edge metrics (PLAND, ED, CA, LPI) capture the proportional extent and spatial dominance of each Urban Structure Type (UST). Shape metrics (SHAPE_MN, SHAPE_SD) describe the geometric complexity and variability of land-cover patches, while aggregation metrics (LSI, AI, NP) quantify landscape fragmentation and clustering patterns. Together, these indices provide a comprehensive representation of urban spatial configuration relevant to land surface temperature dynamics.

All landscape metrics were calculated using the *landscapemetrics* package in R, ensuring standardised, transparent, and reproducible computation (Hesselbarth et al., 2019). Relationships between landscape indices and LST were subsequently examined using correlation analysis to assess how variations in urban form and spatial configuration are associated with surface thermal patterns. This analytical approach is consistent with previous studies demonstrating the importance of landscape structure in urban climate and environmental modelling (Forman, 1995; Liu and Weng, 2009).

2.4. Comparative analysis of LST among USTs

Differences in LST among USTs were assessed using one-way analysis of variance (ANOVA), with UST as the categorical grouping variable representing distinct urban forms and land-use processes. Where relevant, ANOVA assumptions were evaluated using diagnostic checks of residual normality and homogeneity of variances; given the large block-based sample sizes, inference is primarily interpreted in terms of effect patterns across classes rather than strict distributional conformity. Following ANOVA, Tukey's Honestly Significant Difference (HSD) post-hoc test with the Tukey–Kramer adjustment was applied to identify statistically significant pairwise differences between UST classes while accounting for unequal sample sizes (Montgomery, 2017; Salkind, 2010). A stringent significance threshold ($p < 0.001$) was adopted to reduce the likelihood of spurious differences given the large number of comparisons and observations.

2.5. Relationship between landscape metrics and LST

Relationships between landscape metrics and LST were examined using Pearson's correlation coefficient (r) to quantify the strength and direction of linear associations. Correlation significance was assessed using t -tests under standard assumptions of independence and linearity (Cohen et al., 2013). All statistical analyses were conducted in R. Correlation strengths were interpreted as strong ($|r| > 0.7$), moderate ($0.4 \leq |r| \leq 0.7$), or weak ($|r| < 0.4$), with thresholds treated as heuristic guides and interpreted in relation to urban form and thermal processes rather than as absolute criteria (Wilkinson and Friendly, 2009).

To complement the bivariate Pearson correlation analysis, Random Forest regression was used to assess the relative importance of landscape metrics in explaining LST variation within each UST. Random Forest is an ensemble learning method that combines multiple decision trees and is well suited to modelling non-linear relationships and interaction effects among predictors without requiring a predefined functional form (Breiman, 2001). It has been widely applied in remote sensing because of its robustness, predictive performance, and ability to estimate variable importance (Belgiu and Drăguț, 2016), including in recent studies examining the relationship between urban morphology and LST (Han, 2023). Separate Random Forest models were fitted for each UST, with LST as the response variable and landscape metrics as predictors, including PLAND, CA, ED, LPI, NP, LSI, AREA_MN, SHAPE_MN, and SHAPE_SD (as described in Section 2.3). The analysis was conducted in R using the *randomForest* package (Liaw and Wiener, 2002). Each model was fitted with 500 trees, using the default *mtry* setting for regression. Variable importance was evaluated using the permutation-based percentage increase in mean squared error (%IncMSE), where higher values indicate a greater reduction in model accuracy when a predictor is randomly permuted and therefore a greater relative contribution to prediction. The Random Forest analysis is interpreted as a complementary exploratory assessment of typology-specific landscape–LST associations, rather than as evidence of causal effects or directionality. Directional interpretation is based primarily on the Pearson correlation results, while Random Forest is used to strengthen the assessment of relative predictor importance and to identify whether different landscape metrics matter across different

Table 1
Landscape metrics used in the analysis and their mathematical definitions.

Metric group	Metric	Description	Equation	Unit
Area and edge metrics	PLAND	Percentage of landscape occupied by class i	$PLAND_i = (\sum a_{ij} / A) \times 100$	%
	CA	Total area of class i	$CA_i = \sum a_{ij}$	ha
	ED	Edge density of class i	$ED_i = \sum e_{ik} / A$	m/ha
	LPI	Largest patch index of class i	$LPI_i = (\max(a_{ij}) / A) \times 100$	%
Shape metrics	SHAPE_MN	Mean shape index of class i	$SHAPE_MN_i = (1 / n) \sum (0.25 \times p_{ij} / \sqrt{a_{ij}})$	–
	SHAPE_SD	Standard deviation of shape index	$SHAPE_SD_i = \sqrt{[\sum (SI_{ij} - \bar{SI}_i)^2 / (n - 1)]}$	–
Aggregation metrics	LSI	Landscape shape index of class i	$LSI_i = E_i / E(\min)$	–
	AI	Aggregation index of class i	$AI_i = (g_{ii} / g_{ii}(\max)) \times 100$	%
	NP	Number of patches of class i	$NP_i = n$	–

Notes: a_{ij} = area of patch j of class i ; p_{ij} = perimeter of patch j of class i ; A = total landscape area; e_{ik} = length of edge segment k involving class i ; n = number of patches of class i ; E_i = total edge length of class i ; $E(\min)$ = minimum possible edge length for class i ; g_{ii} = number of like adjacencies between pixels of class i ; $g_{ii}(\max)$ = maximum possible like adjacencies; SI_{ij} = shape index of patch j of class i ; $\bar{SI}_i$ = mean shape index of class i .

urban morphologies.

3. Results

3.1. UST spatial configuration

Fig. 2 shows the dominant UST transition within each 500×500 m grid cell between 2010 and 2020, based on the detailed UST maps presented in Fig S1, and indicates substantial shifts in urban form and land-cover composition over the decade. In 2010, agricultural land, forested areas, and other low-intensity land uses remained extensive in the northern and western districts, while compact built-up USTs, such as shophouse typologies, residential areas, and commercial zones, were largely concentrated in the historic urban core and along major development corridors. By 2020, USTs had formed a more continuous, outwardly expanding urban footprint. Built-up typologies extended into formerly peri-urban and semi-rural zones, accompanied by visible contraction of agricultural and vegetated classes along the urban fringe. The urban core shows increased consolidation and diversification of built forms, while peri-urban districts display substantial infill and maturation of earlier development fronts. Areas mapped as “under construction” in 2010

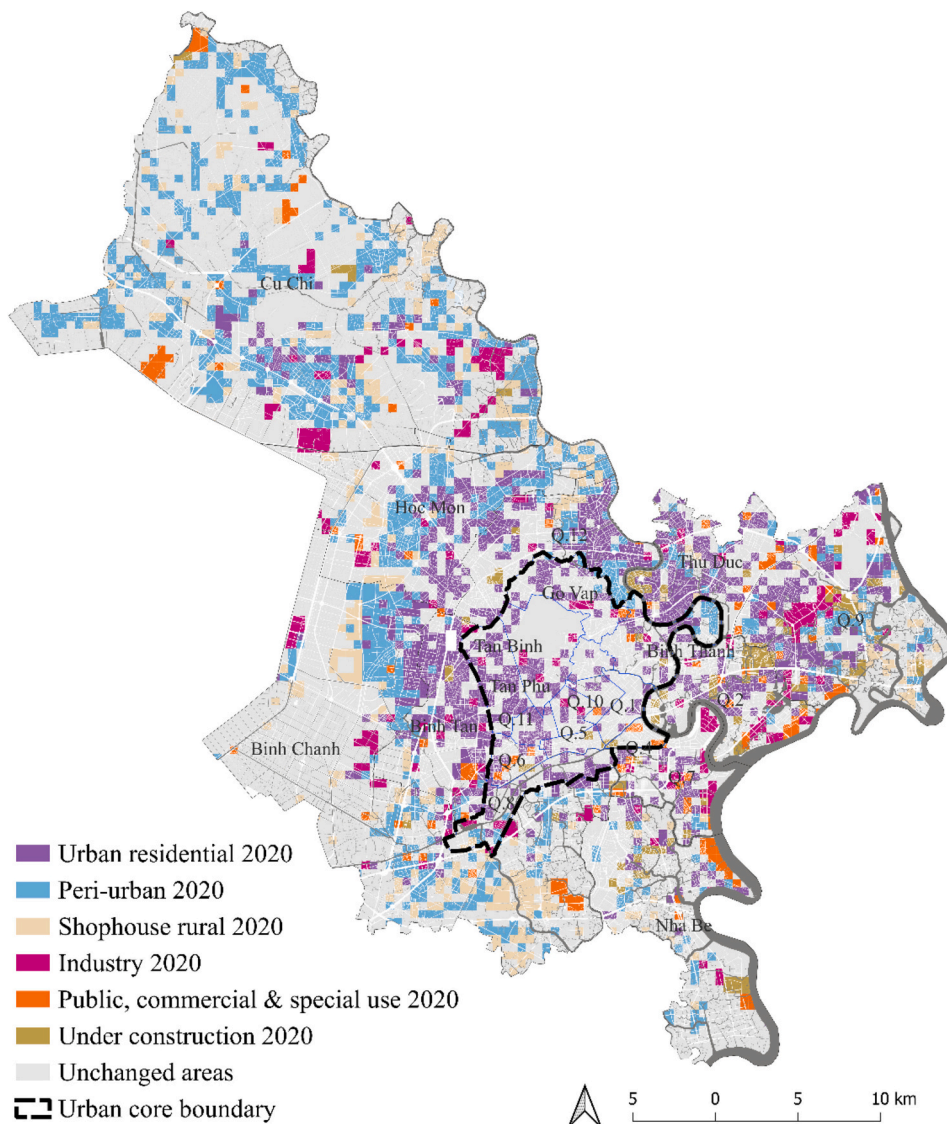


Fig. 2. Spatial development of Urban Structure Types (USTs) in Ho Chi Minh City from 2010 to 2020, visualised by a 500×500 m grid. The outlined area indicates the city's urban core. Each grid cell is classified into one dominant UST transition category for visual interpretation. The urban core and rural areas experienced comparatively limited structural changes. Substantial urban residential expansion was concentrated in the surrounding inner suburban districts, whereas peri-urban development is mainly distributed around the urban fringe, and industrial development is spatially clustered along major transportation corridors and in suburban districts.

largely transitioned into established residential, commercial, or mixed-use USTs by 2020, reflecting the consolidation of previous expansion phases.

3.2. Changes in LST patterns

The transformations in USTs are accompanied by a corresponding increase in dry-season surface temperatures across HCMC. LST shows a clear increase across HCMC between 2010 and 2020, with pronounced intra-urban contrasts (Fig. 3). Overall, warmer conditions concentrated in the dense urban core and major industrial and rapidly urbanising districts, while peripheral districts with greater agricultural and green cover remained comparatively cooler.

Fig. 3 indicates that the highly urbanised central districts (e.g., District 1 and District 3) warmed over the decade, with mean dry-season LST rising from roughly 28–29 °C in 2010 to 30–31 °C in 2020. Industrial and rapidly transforming districts exhibited the strongest increases: Binh Tan and Thu Duc rose from around 29–30 °C in 2010 to 31–32 °C in 2020.

In contrast, peripheral and semi-rural districts recorded lower LST in both periods and smaller absolute changes. Cu Chi and Hoc Mon were around 24–26 °C in 2010, increasing to 26–27 °C by 2020. Several peri-urban districts emerged as new areas of elevated LST. District 7 and parts of Binh Chanh saw temperatures increase from about 28 °C in 2010 to nearly 30 °C in 2020. Previously cooler districts, including Tan Phu and Go Vap also warmed substantially, from around 27 °C to 29–30 °C over the decade.

Overall, the results show a citywide upward shift in dry-season LST, accompanied by a spatial expansion of higher-temperature areas from the historical core toward industrial corridors and peri-urban growth zones. Notably, district-mean LST masks substantial within-district heterogeneity: in several rapidly transforming districts, high temperatures are disproportionately concentrated in industrial, under-construction, and dense shophouse areas, indicating that hotspot formation is closely tied to specific morphologies rather than district averages alone.

3.3. Variations in LST across USTs

LST exhibits pronounced and persistent variation across USTs in HCMC, reflecting systematic differences in built form and land-use function. Table 2 reports the mean LST by UST, and Fig. 4 shows the corresponding contrasts between 2010 and 2020.

Across both time points, compact and heavily built-up USTs are consistently associated with higher surface temperatures than vegetation- and water-dominated land covers. Industrial areas exhibit the highest mean LST (28.91 °C), followed by irregular, mixed, and regular shophouse typologies, as well as public and commercial land uses. These USTs are characterised by dense built form, extensive impervious surfaces, and limited vegetative cover, which collectively are associated with elevated surface temperatures. By contrast, the CBD's pooled mean LST is lower than several dense low-rise fabrics, consistent with daytime shading in taller canyon-like forms. Aquaculture (24.76 °C), forest (25.24 °C), and cropland (25.68 °C) exhibit the lowest mean LST, indicating cooling associated with water- and vegetation-dominated land covers. Notably, green and open spaces exhibit relatively high mean LST, likely because

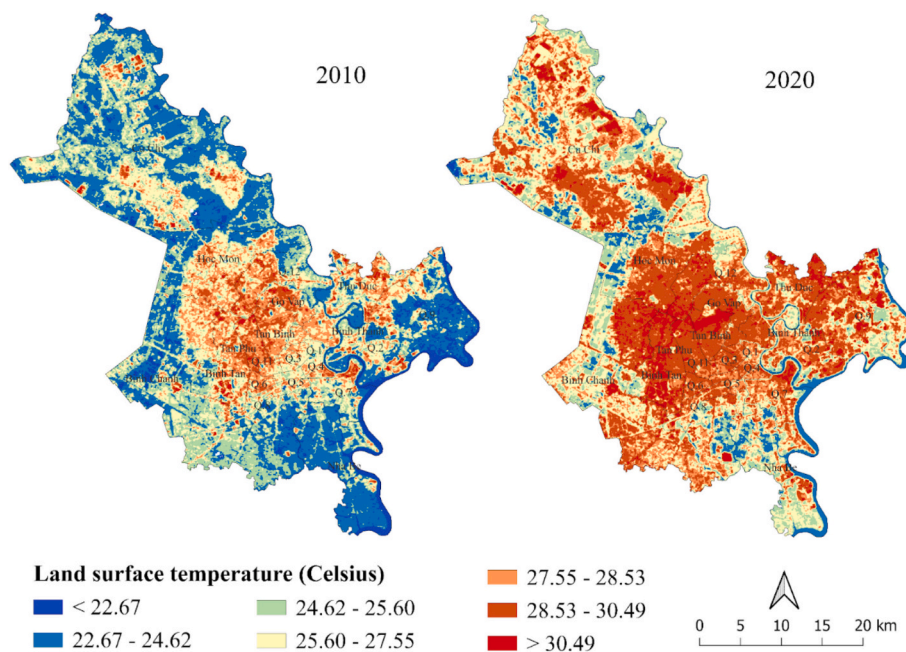


Fig. 3. Dry-season land surface temperature in Ho Chi Minh City in 2010 and 2020, showing a citywide increase in surface temperatures, with pronounced warming in dense urban, industrial, and rapidly urbanising peri-urban areas.

Table 2

Mean land surface temperature by urban structure type (2010 and 2020).

Urban structure type	Mean LST (°C)	n (UST blocks)
Industry	28.91 ± 1.24	6684
Irregular shophouse	28.49 ± 0.78	35,466
Public and commercial zones	28.33 ± 0.85	10,997
Regular shophouse	28.16 ± 0.86	23,276
Mixed shophouse	28.17 ± 1.05	3508
Apartment	28.04 ± 0.88	1938
Green and open spaces	28.03 ± 1.24	2887
Central business district (CBD)	27.59 ± 0.62	2062
Peri-urban shophouse (clustered)	27.67 ± 1.08	8658
Peri-urban shophouse (scattered)	27.13 ± 1.06	5978
Villa	27.22 ± 0.85	2105
Shophouse rural	26.19 ± 1.18	9162
Cropland	25.68 ± 1.57	24,589
Forest	25.24 ± 1.44	1975
Under construction	27.17 ± 1.46	4919
Aquaculture	24.76 ± 1.00	1505

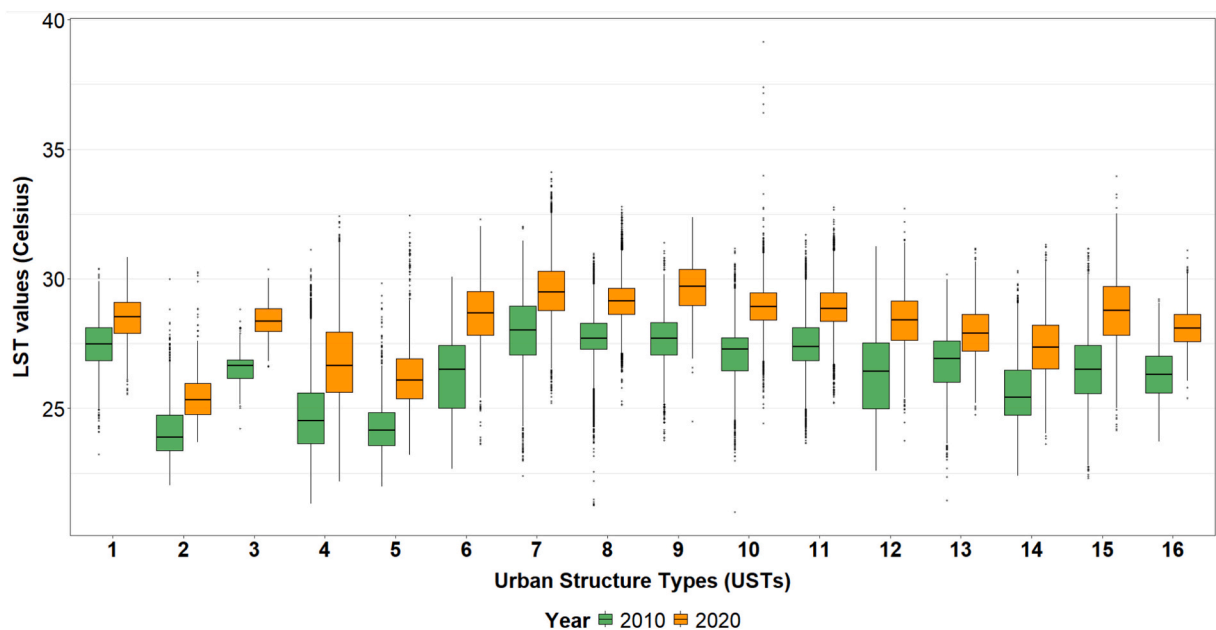


Fig. 4. Land surface temperature (LST) across Urban Structure Types (USTs) in Ho Chi Minh City for 2010 and 2020. *Note:* UST categories comprise: (1) apartment; (2) aquaculture; (3) central business district (CBD); (4) cropland; (5) forest; (6) green and open spaces; (7) industry; (8) irregular shophouse; (9) mixed shophouse; (10) public, commercial and special uses; (11) regular shophouse; (12) peri-urban shophouse (clustered); (13) peri-urban shophouse (scattered); (14) shophouse rural; (15) under construction; and (16) villa. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article)

Table 3

Relationship between changes in UST extent and changes in LST (2010–2020).

Urban structure type	β ($\Delta\text{LST} \sim \Delta\text{UST area}$)	p-value	Direction
Industry	6.1×10^{-6}	<0.001	Warming
Public and commercial zones	4.4×10^{-6}	<0.001	Warming
Regular shophouse	4.5×10^{-6}	<0.001	Warming
Forest	-5.9×10^{-6}	<0.001	Cooling
Cropland	-3.5×10^{-6}	<0.001	Cooling
Aquaculture	-1.1×10^{-6}	n.s.	Neutral

Note: Table 3 reports a subset of USTs for brevity. Shown are USTs with statistically significant associations ($p < 0.001$), plus aquaculture as a non-significant reference. Full coefficients for all 16 USTs are reported in Table S1. $\Delta\text{UST area}$ was measured within 500×500 m grid cells.

this class includes a mix of canopy-poor open areas (e.g., lawns/sports grounds, paved plazas, vacant or bare ground) that can heat strongly during the dry season despite being ‘open’. One-way ANOVA and Tukey’s HSD post hoc tests confirm that differences in mean LST among USTs are statistically significant ($p < 0.001$; Table 2). Full statistical outputs, including complete Tukey HSD groupings and diagnostic checks, are provided in Supplementary Table S1. Although LST increased across all USTs between 2010 and 2020 (Fig. 4), this cross-year warming signal should be interpreted cautiously given potential cross-sensor and processing differences; nevertheless, the relative ranking of UST thermal intensity remains broadly similar in both years, suggesting persistent morphology-related spatial heat contrasts.

Table 3 extends this comparison by examining the association between changes in UST extent and changes in LST from 2010 to 2020. Expansion of industrial, public-commercial, and regular shophouse USTs is significantly associated with increases in Δ LST. In contrast, the cooling effects of forests and croplands are significantly diminished in proportion to the loss of their area. While effect sizes are modest (β coefficients reflect Δ LST per unit change in UST area), the consistency and statistical significance of these relationships indicate cumulative thermal impacts of incremental urban structural change.

3.4. Relationship between LST and landscape structure

Relationships between LST and landscape structure vary across USTs, with distinct patterns in the strength and sign of specific metrics (Table 4). Across most USTs, PLAND shows positive associations with LST, with the strongest correlations observed for transformation-related and dense built-up classes, including under construction ($r = 0.57$) and shophouse typologies ($r = 0.45$ – 0.53). Industrial and public-commercial areas show similarly positive relationships ($r = 0.44$), indicating higher LST where these classes occupy a larger share of the landscape. Other composition and dominance metrics (e.g., CA and LPI) show more UST-specific behaviour, including negative correlations in several non-built classes and some peri-urban forms, but weak or positive associations in dense urban typologies. Configuration metrics (ED and NP) exhibit mixed correlations with LST, suggesting that fragmentation and patchiness relate to thermal conditions in ways that depend on the underlying urban form and land-cover context. By comparison, SHAPE_MN is consistently negative across USTs ($r \approx -0.13$ to -0.62), indicating that greater patch-shape complexity tends to coincide with lower LST, although the strength of this relationship varies across urban forms. Overall, Table 4 suggests that LST patterns are most consistently associated with landscape composition (especially class proportions) and patch-shape characteristics. In contrast, relationships with fragmentation-related metrics are more context-dependent across USTs.

The Random Forest analysis confirms that landscape–LST relationships are typology-specific rather than governed by a single universal metric (Fig. 5; Table S3). Across USTs, SHAPE_MN, PLAND, Area_MN, LSI, and ED frequently emerge as important predictors, but their relative importance varies strongly by urban morphology. PLAND is especially important in several dense built-up and shophouse-related USTs, including Industry, Irregular shophouse, Regular shophouse, and Shophouse rural, indicating the role of class dominance in structuring surface heat. By contrast, SHAPE_MN is the leading predictor in several residential, green/open, peri-urban,

Table 4

Pearson correlation coefficients (r) between land surface temperature (LST) and selected landscape metrics by Urban Structure Type (UST).

Urban Structure Type	PLAND	CA	ED	LPI	NP	SHAPE_MN
Apartment	0.30	−0.15	−0.20	−0.25	−0.15	−0.52
Aquaculture	0.33	−0.08	−0.10	−0.10	−0.08	−0.25
CBD	−0.04	−0.47	−0.41	−0.17	−0.08	−0.13
Cropland	0.24	−0.27	−0.23	−0.14	−0.13	−0.32
Forest	0.35	−0.22	−0.24	−0.14	−0.20	−0.36
Green & open space	0.37	−0.17	−0.10	−0.20	0.08	−0.18
Industry	0.44	0.16	0.19	0.01	0.21	−0.45
Irregular shophouse	0.53	0.24	0.14	−0.01	0.05	−0.55
Mixed shophouse	0.45	0.07	−0.03	−0.01	−0.14	−0.48
Peri-urban clustered	0.44	0.16	0.12	−0.21	0.08	−0.51
Peri-urban scattered	0.44	−0.06	−0.07	−0.02	−0.07	−0.46
Public & commercial	0.44	−0.01	0.00	−0.04	0.07	−0.32
Regular shophouse	0.52	0.05	−0.02	0.06	−0.08	−0.51
Shophouse rural	0.48	−0.10	−0.12	0.05	−0.11	−0.44
Under construction	0.57	−0.01	−0.11	0.19	−0.12	−0.20
Villa	0.41	−0.19	−0.23	−0.15	−0.20	−0.62

Notes: PLAND = percentage of landscape; CA = class area; ED = edge density; LPI = largest patch index; NP = number of patches; SHAPE_MN = mean shape index. Statistical significance varies across metrics and USTs; full correlation coefficients and significance levels are provided in Supplementary Table S2. Cells shaded in grey indicate moderate-to-strong correlations ($|r| \geq 0.40$).

and transitional morphologies, suggesting that patch geometry and spatial complexity also influence LST patterns. These results support the Pearson correlation analysis while adding further evidence that both landscape composition and configuration shape surface heat differently across urban forms.

4. Discussion

4.1. Urban morphology as a structuring mechanism of surface heat in HCMC

This study shows that surface temperature in HCMC is not adequately explained by generic “urbanisation” or land-cover change alone. Rather, dry-season LST is systematically structured by urban morphology, captured here through USTs that encode built form, functional use, and the spatial organisation of green–blue elements. In this sense, the city’s thermal landscape is not simply a climatic outcome, but a spatial imprint of how development is assembled, consolidated, and governed over time (Liu and Weng, 2009; Stewart and Oke, 2012; Voogt and Oke, 2003).

At the district scale, HCMC shows an apparent upward shift in dry-season LST between 2010 and 2020, with warming concentrated in the historic core and expanding into industrial corridors and peri-urban growth districts (Section 3.2). The outward diffusion of higher LST signals that higher surface-heat exposure potential is increasingly emerging at the urban fringe, where infrastructure sequencing, planning control, and green–blue provision often lag behind development pressures (Downes et al., 2024; Duy et al., 2025; Scheiber et al., 2025).

These patterns are consistent with earlier studies of HCMC, which show that rapid built-up expansion, industrial development, and densification have contributed to higher surface temperatures, particularly in dense urban and peri-urban growth areas (Doan et al., 2016; Son et al., 2017; Katschnner et al., 2023). Similar relationships have been reported across Southeast Asian megacities, where conversion of vegetated and agricultural land into impervious urban surfaces has been associated with stronger surface urban heat

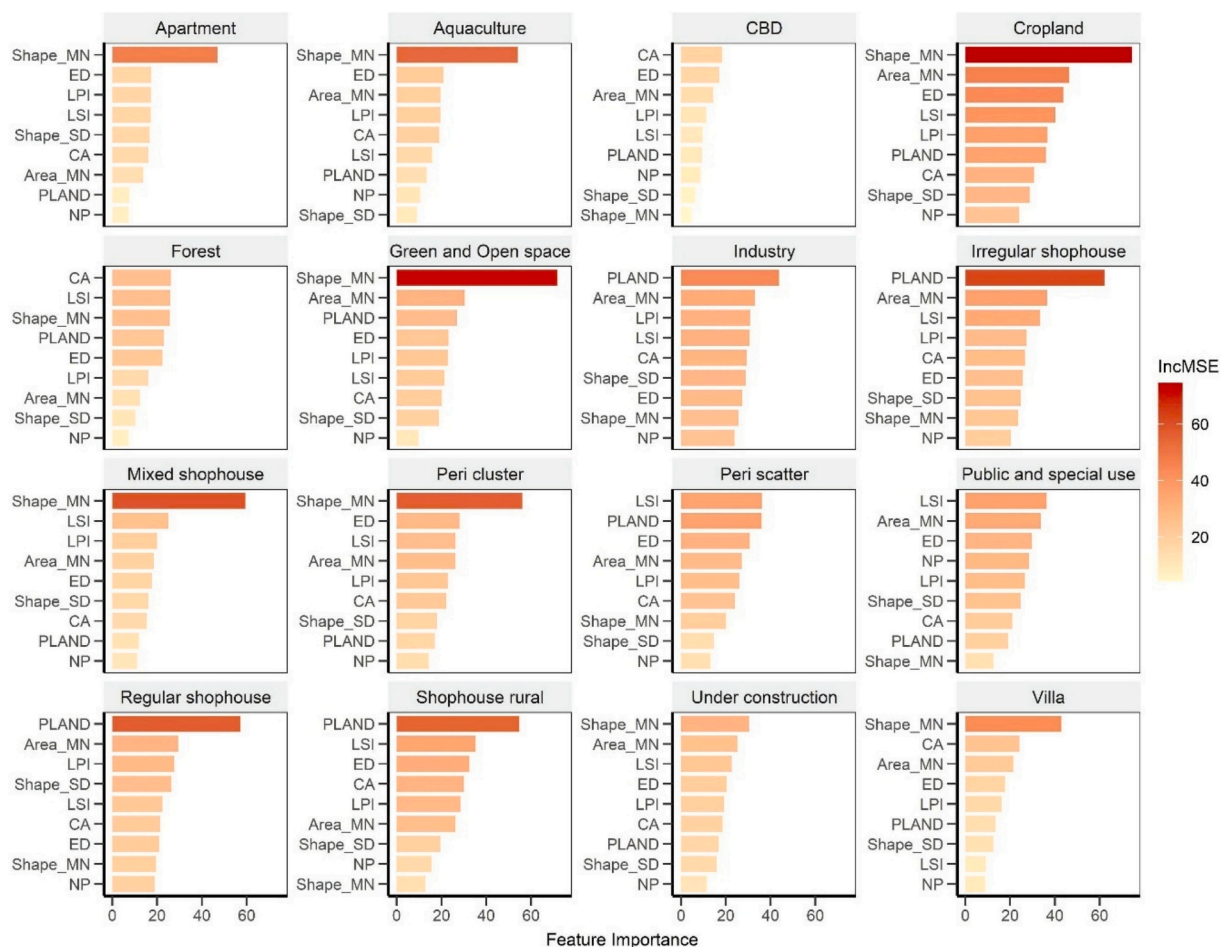


Fig. 5. Random Forest variable importance of landscape metrics for explaining land surface temperature within each Urban Structure Type. Variable importance is expressed as a percentage increase in mean squared error (%IncMSE), where higher values indicate a greater contribution of the metric to the model's prediction accuracy.

island effects (Estoque et al., 2017; Nguyen et al., 2023). Our findings extend this work by showing that the thermal signal is not only linked to urban land-cover change in general, but is differentiated by locally specific USTs, including industrial/logistics areas, dense shophouse fabrics, and peri-urban transition zones.

This contribution is particularly relevant from the perspective of a Southeast Asian megacity. Many cities in the region are experiencing simultaneous densification of historic cores, expansion of industrial corridors, and incremental peri-urban conversion, often under conditions of fragmented development control and uneven provision of green–blue infrastructure. In such settings, LST patterns cannot be fully interpreted through broad urban–rural gradients or land-cover categories alone. The HCMC case shows that surface heat emerges through multiple development pathways: large-scale industrial and logistics landscapes, high-coverage shophouse fabrics, and development-in-motion areas can all produce elevated LST, but through different morphological mechanisms. The UST approach, therefore, adds a planning-relevant layer to existing SUHI and LCZ studies by linking surface thermal conditions to development fabrics that are more directly legible in terms of zoning, design standards, and retrofit strategies (Diem et al., 2024a; Katzschner et al., 2016; Ren et al., 2011; Storch and Downes, 2013).

The UST results sharpen this diagnosis by demonstrating a persistent thermal hierarchy with most morphologies warmer in 2020 (Section 3.3; Table 2; Table S1). It indicates that morphological configuration acts as a durable structuring mechanism: it conditions where heat accumulates, how it persists, and how far it extends through adjacent surfaces as the city expands and becomes more compact.

This distinction matters because much remote-sensing work in HCMC documents convincing biophysical correlations between built-up expansion and increases in LST, yet commonly treats the city as a surface mosaic of land-cover classes and indices (Thien et al., 2025). Such approaches tend to under-specify the development pathways that produce those surfaces, the planning regimes, construction logics, and incremental transformations that create distinct morphologies and constrain future retrofits. A UST lens enables the interpretation of heat patterns produced through form–function reconfiguration rather than mapping them onto a static land-cover baseline. This aligns with wider syntheses arguing that urban heat is an emergent property of “*form, function, and fabric*”, and that interventions targeting only surface materials or greening “*fabric*” can leave deeper morphological lock-in intact (Eisenstein, 2025; Santamouris, 2015).

The grid-based change model reinforces this claim of “surface heat production through structural change” (Table 3; Table S1). Expansion of various USTs within built-up areas, including Industry, Public/Commercial/Special Use, Regular Shophouse, and Under Construction areas, is differentially associated with increases in Δ LST. These coefficients imply meaningful cumulative warming when scaled to district-level expansion, consistent with the thermal consequences of incremental land conversion during rapid urban growth. Importantly, some already-hot morphologies are not the strongest statistical drivers of change. For instance, Irregular Shophouse exhibits a high mean LST but a comparatively weak/non-significant expansion–warming signal ($p \approx 0.053$).

4.2. Landscape composition, configuration, and typology-specific LST relationships

Landscape-metrics results (Section 3.4; Table 4; Table S2) clarify how urban morphology is associated with surface thermal conditions across USTs. Across most built-up USTs, LST is most consistently related to composition and dominance, particularly PLAND. This finding is consistent with previous studies showing that built-up intensity, impervious surface dominance, and reduced vegetation cover are strongly associated with elevated LST in urban landscapes (Liu and Weng, 2009; Yue et al., 2019; Rahimi et al., 2021). Comparative evidence across Vietnamese cities similarly shows that surface temperature patterns relate systematically to building density and green-space cover, reinforcing the relevance of composition metrics beyond a single-city case (Braun et al., 2023).

In HCMC, composition signals are strongest in transformation-related and dense built forms, including Under Construction, Irregular Shophouse, Regular Shophouse, Industry, and Public/Commercial/Special Use. By contrast, fragmentation metrics such as ED and NP show more mixed and context-dependent relationships, suggesting that they should not be interpreted as universal planning levers. This aligns with studies indicating that landscape–LST relationships are scale-sensitive and vary according to the spatial embedding of built-up, vegetated, and water surfaces (Dugord et al., 2014; Rahimi et al., 2021; Jeon et al., 2023). One consistent pattern is that SHAPE_MN is negative across all USTs, suggesting that greater patch-shape complexity tends to coincide with lower LST, potentially reflecting more porous or interdigitated configurations in which built fabric is interrupted by vegetated edges, water lines, or heterogeneous open space.

The Random Forest variable-importance analysis strengthens this interpretation by showing that landscape–LST relationships are typology-specific rather than governed by a single dominant metric (Fig. 5; Table S3). PLAND emerges as an important predictor in several dense built-up and shophouse-related USTs, supporting the role of class dominance in structuring surface heat. However, SHAPE_MN, LSI, Area_MN, and ED are also important in several open, peri-urban, and mixed morphologies, indicating that configuration and patch geometry matter differently depending on urban form. Together, the Pearson correlation and Random Forest results suggest that surface heat mitigation priorities should differ by typology: reducing the areal dominance of heat-intensive classes and increasing vegetated or permeable fractions within them appears more consistently relevant than targeting fragmentation metrics in isolation.

4.3. Planning and governance implications

In planning terms, composition metrics more reliably indicate where surface thermal burdens are structurally concentrated, while configuration metrics help explain how cooling capacity may be mediated by connectivity, edges, and the porosity of the built fabric. These findings align with multi-scale evidence that built-up dominance metrics are often strongly associated with LST change, and that

these relationships can strengthen with scale and exhibit spatial spillover effects (Han, 2023; Meng et al., 2022; Rahimi et al., 2021). For HCMC, this suggests that individually small land-conversion processes may add up to district-scale surface thermal shifts, particularly in peri-urban growth zones where outward warming is linked to the cumulative expansion and consolidation of heat-intensive urban fabrics.

The results suggest that surface-heat mitigation in HCMC should move beyond generic greening or citywide cooling strategies toward typology-specific interventions. The UST framework provides a practical intermediate scale between broad land-cover classes and highly detailed parcel-level design: it captures forms that planners and developers recognise, such as industrial/logistics areas, dense shophouse fabrics, public-commercial areas, under-construction zones, and peri-urban transition landscapes. Because these USTs exhibit distinct LST profiles and distinct landscape-metric relationships, they require different mitigation priorities. This aligns with broader evidence that urban cooling is rarely achieved by single interventions alone, but by combined measures across surface materials, vegetation, shading, ventilation, and neighbourhood form (Santamouris, 2015; Yang et al., 2025).

These findings do not imply a single greening solution for HCMC. Rather, they suggest different intervention logics for different hotspot-producing fabrics. In dense shophouse areas, where elevated LST is associated with high impervious coverage, narrow street sections, and limited vegetated relief, mitigation should prioritise street shading, pocket greening, courtyard retrofits, façade shading, cool roofs, and protection of remaining permeable surfaces, consistent with microclimate evidence from HCMC shophouse environments (Dang and Pitts, 2020). In industrial and logistics areas, where large sealed surfaces and extensive roof areas contribute to high surface temperatures, priorities include cool or reflective roofing, shaded loading areas, tree planting along internal roads, surface-water retention, and minimum vegetated or permeable fractions. In under-construction and peri-urban transition zones, the most important opportunity is preventive rather than corrective: green-blue corridors, plantable street sections, plot-coverage limits, and shade standards should be embedded before urban form becomes fixed.

This distinction is important because HCMC's peri-urban warming cannot be reduced to a simple "sprawl" narrative or a density gradient radiating from the centre. The 2010–2020 transition reflects the consolidation of multiple development pathways, including planned corridor growth, industrial expansion, and incremental plot-by-plot densification. Both planned and organic morphologies can produce elevated LST, but through different mechanisms: planned large-scale developments may generate extensive sealed surfaces and delayed canopy establishment, while irregular shophouse fabrics often intensify surface heat through high plot coverage, narrow streets, and limited space for vegetation. The planning challenge is therefore not to treat one development pathway as inherently cooler than another, but to identify the specific surface-heat signature of each morphology and intervene accordingly.

A further planning implication concerns potential surface-heat lock-in at the urban fringe. The results indicate that elevated LST is increasingly associated with peri-urban consolidation and corridor growth, where long-lived morphological conditions are established early through plot coverage, sealed surfaces, street sections, plantable surfaces, and the continuity or loss of green-blue corridors. Because these features are difficult and costly to reverse once built, surface-heat mitigation options may become progressively constrained over time, particularly where industrial land conversion and high-coverage housing types dominate new development fronts. In this sense, the UST framework provides a planning translation layer between satellite-derived surface-heat diagnostics and morphology-specific mitigation options.

4.4. Limitations and directions for future research

A key limitation is the limited inter-temporal comparability of LST, because the 2010 and 2020 estimates are based on different Landsat missions (Landsat 5 TM vs Landsat 8 OLI/TIRS). Cross-sensor differences in radiometry, thermal band characteristics, and retrieval assumptions can shift absolute LST values even where relative spatial patterns remain robust, so results are best interpreted as morphology-consistent signals of thermal change rather than sensor-invariant point estimates (Morfitt et al., 2015; Sekertekin and Bonafoni, 2020). Uncertainty may also arise from compositing and atmospheric/emissivity choices, which can influence absolute LST magnitudes in humid, cloud-affected settings; future work should therefore report sensitivity tests across plausible compositing/retrieval parameterisations and, where possible, apply cross-sensor harmonisation or normalisation approaches. Landsat thermal bands are natively coarser and commonly resampled to 30 m, so fine-scale canyon effects are not resolved; higher-resolution thermal sources (airborne/UAV or emerging satellite products) would strengthen street-scale interpretation and targeting (Nguyen, 2026). UST classification itself introduces uncertainty in rapidly transforming landscapes: field validation and patch attribution are time-intensive, and misclassification may affect UST–LST contrasts, motivating expanded ground-truthing and reproducible, updateable classification pipelines. Although focusing on dry-season imagery reduces seasonal confounding, inter-annual climate variability can still influence LST independently of land conversion, so future studies should incorporate multi-seasonal analysis and basic climate covariates to better separate land-cover/morphology effects from background conditions. Finally, because temperature fields and urban form are spatially clustered, statistical relationships may be influenced by spatial dependence; follow-on work should test sensitivity using spatial diagnostics and spatially robust inference to support planning interpretation by appropriately cautious claims. Substantively, dry-season daytime LST captures the surface thermal imprint of urbanisation rather than human heat stress, thermal comfort, or health risk. Because this study does not include population exposure, socio-economic vulnerability, adaptive capacity, indoor thermal conditions, or behavioural responses, it should not be interpreted as a full heat-risk assessment. In HCMC, long-term station-based observations show that the heat index typically peaks in April–May and has increased over 1990–2023, reinforcing the need to connect satellite-derived LST diagnostics to human-relevant heat metrics (Vo and Nguyen, 2026). Future research should assess how morphology–temperature relationships vary across different periods of the day, including daytime, evening, and nighttime conditions, using complementary satellite observations and ground-based monitoring. Such work should combine UST-based LST diagnostics with near-surface air temperature, humidity, indoor overheating, social vulnerability indicators, and perceived thermal comfort, especially

in dense, low-ventilation fabrics where elevated surface heat and limited adaptive capacity may overlap (Bardhan et al., 2024; Dang and Pitts, 2020; Franck et al., 2013; Sun et al., 2025).

5. Conclusions

This study shows that surface temperature in HCMC is not adequately described by generic outcomes of urbanisation or land-cover change alone. Rather, dry-season Landsat LST is systematically structured by urban morphology, captured here through USTs that encode built form, function, and the spatial organisation of green–blue elements. Between 2010 and 2020, HCMC experienced an apparent citywide upward shift in LST, with higher-temperature areas expanding from the historic core into industrial corridors and peri-urban growth zones. This indicates that surface heat is increasingly associated with the consolidation of specific urban development fabrics at the urban fringe.

Across the city, USTs exhibit a persistent thermal hierarchy. Industrial/logistics areas, commercial–service fabrics, dense shop-house typologies, and under-construction or transition zones remain consistently warmer than vegetation- and water-dominated morphologies. Change analysis further indicates that the expansion of industrial, commercial/service, and development-in-motion classes aligns with increases in Δ LST, whereas the retention of forest and cropland is associated with cooler surface conditions. These findings support an interpretation of surface heat as structured by specific development trajectories rather than distributed evenly across urban space.

Landscape-metric analysis reinforces a practical planning signal. Across built-up USTs, LST relates most consistently to areal dominance and composition, especially PLAND, CA, and LPI, whereas configuration metrics are more mixed and context-dependent. The Random Forest variable-importance analysis further shows that landscape–LST relationships are typology-specific, with different combinations of composition and configuration metrics mattering across different urban morphologies. This suggests that surface-heat mitigation priorities should be defined by morphology: limiting the dominance of hotspot-producing fabrics, increasing vegetated and permeable fractions within them, and preserving green–blue continuity before development patterns become fixed.

Finally, LST should be interpreted as the surface thermal imprint of land conversion and morphology, not as heat risk in itself. Human heat impacts also depend on near-surface air temperature, humidity, radiation, nocturnal relief, indoor conditions, exposure, socio-economic vulnerability, and adaptive capacity. The contribution of this study is therefore to identify how urban development structures surface heat-exposure potential and creates potential surface-heat lock-in through plot coverage, street sections, shadeable and plantable surfaces, and green–blue continuity. USTs provide an operational bridge between satellite-derived thermal diagnostics and planning instruments, supporting typology-specific surface-heat mitigation in rapidly transforming tropical megacities.

Future research should (i) test sensitivity to spatial dependence using spatial diagnostics and spatially robust inference, (ii) strengthen multi-temporal comparability through cross-sensor harmonisation and compositing sensitivity checks, and (iii) extend the UST approach by linking hotspot grids to indoor/outdoor monitoring, night-time thermal conditions, perceived thermal comfort, and socio-economic vulnerability indicators.

CRediT authorship contribution statement

Nigel K. Downes: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Phan Kieu Diem:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Can Trong Nguyen:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Nguyen Kieu Diem:** Writing – review & editing, Visualization, Methodology, Formal analysis, Data curation. **Tran Gia Hong:** Visualization, Methodology, Formal analysis, Data curation. **Andrea Reimuth:** Writing – review & editing, Writing – original draft.

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Declaration of competing interest

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.uclim.2026.103129>.

Data availability

The authors do not have permission to share data.

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